

Jason Tsao

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EDUCATION

San Jose State University

San Jose, CA

Bachelor of Science in Computer Engineering, Minor in Japanese (GPA 3.70)

Aug. 2024 – Dec. 2027

- **Relevant Coursework:** Data Structures and Algorithms; Object-Oriented Concepts & Methodology; Circuit Analysis; Site Reliability Engineering for Cloud

EXPERIENCE

Systems Administrator

Feb. 2026 – Present

Student Union, Inc. of SJSU

San Jose, CA

- Managed Proxmox VE, VM provisioning, backups, shutdown and high availability for reliable service continuity.
- Implemented Kubernetes for containerized workloads, enabling scalable deployments and operations for 40k users.
- Diagnosed connectivity issues spanning SSL certificates, TCP/IP, DHCP, DNS, VLANs, NAT rules, and firewalls.
- Configured SNMP server to monitor printer status, toner levels, and supplies for asset management and ordering.
- Assisted with MDF/IDF infrastructure support, including drop ports, switches, patching, and punchdowns.
- Administered Windows and macOS endpoints, Windows Server, Active Directory, Microsoft Entra ID, and Intune.
- Managed printers, workstation deployment, imaging, software installation, and patch management.
- Secured endpoints with Qualys, Sophos, Jamf, and Intune and automated tasks with PowerShell scripts.
- Conducted technical interviews to screen and evaluate candidates for IT support roles.
- Coordinated with external vendors and SaaS providers to procure service plans and resolve technical issues.

Software Engineer, Development Team

Jan. 2026 – Present

Software and Computer Engineering Society

San Jose, CA

- Built a YouTube-to-Raspberry-Pi music streaming app in TypeScript with React, Express, Prisma, and SQLite.
- Designed an egress-only WebSocket bridge so the Pi dials out to the backend, removing inbound firewall rules.
- Containerized services with Docker Compose and ran the Pi daemon as a systemd unit piping yt-dlp into mpv.

Operations Supervisor

Sep. 2024 – Feb. 2026

Student Union, Inc. of SJSU

San Jose, CA

- Supervised and developed staff by monitoring performance and providing hands-on coaching as needed.
- Onboarded new hires, enforced permit compliance for organizations, and coordinated event logistics.
- Directed emergency procedures including evacuations, fire panel checks, and resolution of security incidents.

PROJECTS

Home Lab | *Docker, Ansible, Nginx, Prometheus, Grafana, Ollama, Proxmox, pfSense, OpenMediaVault, WireGuard*

- Configured pfSense as the primary router with VLANs, custom firewall rules, and NAT policies.
- Deployed Proxmox as a bare-metal hypervisor to host, isolate, and snapshot multiple homelab VMs.
- Automated host configuration and SSH key distribution with idempotent, agentless Ansible playbooks.
- Deployed OpenMediaVault with a virtual-disk RAID 10 storage pool for fault-tolerant network-attached storage.
- Configured Prometheus and Grafana for real-time metrics collection, alerting, and custom dashboards.
- Self-hosted Ollama for local LLM inference behind Nginx with Cloudflare SSL over a WireGuard VPN.

StudyGuard | *Python, FastAPI, OpenCV, OpenAI Whisper, SQLAlchemy*

- Won **2nd Place** at the Seed Embodied AI Hackathon for building an AI-powered focus-monitoring system.
- Built an OpenCV and GPT-4V pipeline that classifies student focus states at 10-second intervals.
- Streamed live camera feed to a FastAPI dashboard for real-time proctoring and focus visualization.
- Used a Whisper voice agent to control Reachy Mini robot gestures and update the dnsmasq allowlist.

TECHNICAL SKILLS

Languages: C/C++, Assembly, JavaScript, TypeScript, HTML/CSS, Python, Bash/Zsh, PowerShell

Frameworks & Libraries: React.js, Vite, Node.js, Express, FastAPI, Prisma, SQLAlchemy, SQLite

Infrastructure & DevOps: Docker, Proxmox, pfSense, Nginx, Prometheus, Grafana, Ollama, WebSockets, TCP/IP, DHCP, DNS, VLAN, WireGuard, SSL/TLS, Linux, Git, GitHub Actions, CI/CD

Security & Endpoint Management: Qualys, Sophos Central, Jamf, Intune, Active Directory, Windows Server

Certifications: Technical Support Fundamentals (Google); Vulnerability Management Detection and Response (Qualys)